higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MULTIMEDIA SERVICES

NQF LEVEL 4

(10040184)

20 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 9 pages.

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TIME: 3 HOURS
MARKS: 150

INSTRUCTIONS AND INFORMATION

1. Read ALL the questions carefully.
2. Answer ALL the questions.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Write neatly and legibly.

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SECTION A

QUESTION 1

1.1.1 Which ONE of the following is NOT a type of firewall?

A Packet filtering firewall
B Dual homed gateway firewall
C Screen host firewall
D Word firewall

1.1.2 In message confidentiality, the transmitted message must make sense to the intended … only.

A receiver
B sender
C modular
D translator

1.1.3 A digital signature needs a … system.

A network-key
B shared-key
C public-key
D wi-fi key

1.1.4 ONE way to preserve the integrity of a document is by using a/an …

A eye-ray.
B finger print.
C laser rays.
D x-rays.

1.1.5 This type of database stored *objects* rather that *data*:

A Non-relational
B Object-oriented
C Distributed
D Relational

1.1.6 When data should arrive at the receiver exactly as it was sent, it is called message …

A confidentiality.
B integrity.
C splashing.
D sending.

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1.1.7 Why are cyber vulnerabilities unlikely to disappear?

A Cyber vulnerabilities are protected in a secret base on the moon
B Criminals need vulnerabilities to steal identities
C Vulnerabilities are side effects of the freedom and ease of communicating online
D The government would not allow people to fix vulnerabilities

1.1.8 The purpose of a denial of service attack is to …

A exploit a weakness in the TCP/IP stack.
B execute a Trojan virus on a system.
C overload a system to prevent it from being operational.
D shutdown services by turning them off.

1.1.9 … refers to a network's capability to discover unauthorised or unlawful access or unlawful traffic.

A Intrusion prevention
B Worm
C Trojan horse
D Intrusion detection

1.1.10 The most important activity in system hacking is …

A information gathering.
B cracking passwords.
C escalating privileges.
D covering tracks.

1.1.11 Phishing is a form of …

A spamming.
B identify theft.
C impersonation.
D scanning.

1.1.12 The best statement for taking advantage of a weakness in the security of an IT system is called …

A a threat.
B an attack.
C exploitation.
D vulnerability.

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1.1.13 WPA2 is used for security in …

A ethernet.
B bluetooth.
C wi-fi networks.
D 3G networks.

1.1.14 When a DNS server accepts and uses incorrect information from a host that has no authority to give that information, it is called …

A DNS lookup.
B DNS hijacking.
C DNS spoofing.
D denial-of-service.

1.1.15 A web cookie is a small piece of data that is …

A sent from a website and stored in a user's web browser while a user is browsing a website.
B sent from a user and stored in the server while a user is browsing a website.
C sent from the root server to all servers.
D automatically created by a client computer for browsing.

(15 × 1) (15)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.2.1–1.2.10) in the ANSWER BOOK.

1.2.1 A representation of the structure of a database is called a schema.

1.2.2 A cache is used to maximise the amount of physical disk input and output performed by a DBMS.

1.2.3 SQL is case-*insensitive* (but data is case-*sensitive*).

1.2.4 UNITES are used to combine data from two tables.

1.2.5 A database should always contain redundant information.

1.2.6 *Insert* and *delete* are SQL commands used to create and delete tables.

1.2.7 A foreign key refers to ONE or more values in a database table that identify records in another table.

1.2.8 A network administrator is responsible for the design, control and administration of a relational database.

1.2.9 The internet of today is heavily dependent on databases.

1.2.10 A database consists of multiple records that belong together.

(10 × 1) (10)
[25]

TOTAL SECTION A: 25

SECTION B

QUESTION 2

2.1 Briefly explain how the following factors affect the performance of a TCP/IP network.

2.1.1 Network errors

2.1.2 Distance

2.1.3 Memory (RAM)

2.1.4 Firewalls and filters

(4 × 2) (8)

2.2 Describe the functions of a DHCP server in a WAN setup. (3)

2.3 Tabulate FIVE differences between desktop operating systems and server operating systems. (5 × 2) (10)

2.4 Identify FOUR server operating system software. (4)
[25]

QUESTION 3

3.1 Explain TWO benefits of using passwords on servers. (2 × 2) (4)

3.2 With the aid of a diagram explain hierarchical database structure. (8)

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3.3 Study the table below and answer the questions.
Database name: Staff

Employee Table

ID	First Name	Last Name	Address	City	Age
101	Mickeys	Mousy	223 Fanta Ave	Durban	56
102	Batty	Weli	111 Press St	Durban	25
103	Winder	Petty	89 Honest Way	Pretoria	76
104	Donald	Dicky	45 Quiz Street	Soweto	45
105	Benny	Bulls	41 Achy Str	Brits	35
106	Widza	Ndoro	1 Atoms Road	George	54
107	Caty	Nelson	8 Perfect Street	Pretoria	22
108	Enos	Mthembu	100 Life Ave	George	56
109	Amon	Azulu	124 Owl Road	Brits	44
110	Elsie	Peterse	4 King Ave	Durban	61

Create SQL queries for the following:

3.3.1 Display all employees who are older than 50 years and live in Durban. (3)

3.3.2 INSERT a new employee into the table with the following details: (ID = 112; First Name = Bill; Last Name = Baloyi; Address = 20 Upper Rd; City = Brits; Age = 48). (4)

3.3.3 Display ALL the records in the table. (2)

3.3.4 Delete the record of Batty Weli from the table. (2)

3.3.5 Identify the column that could be used as the primary key in the table. Give ONE reason for your selection. (2)
[25]

QUESTION 4

4.1 Differentiate between a traditional *Peer-to-Peer* network and a *P2P* network used on website such as Kazaa and BirTorrent. (7)

4.2 Describe the evolution of Web 2.0. (4)

4.3 Create a JavaScript function that will accept any two numbers from the user and return the sum. (12)

4.4 Briefly explain the term *Web - or Internet freedom.* (2)
[25]

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QUESTION 5

5.1 You intend to create your own personal website to market your own products.

How could the following help you come up with a good design:

5.1.1 Defining your target audiences

5.1.2 Setting your goals and objectives

5.1.3 Creating a sitemap

5.1.4 Setting a budget for your website

(4 × 2) (8)

5.2 Determine the outcome of the following JavaScript statement:

```
var i = 20;

if (i == 10)
  document.write("i is 10");
else if (i == 15)
  document.write("i is 15");
else if (i == 20)
  document.write("i is 20");
else
  document.write("i is not present");
< /script>
```

(2)

5.3 Discuss where the following intrusion detection systems should be placed to protect an enterprise network:

5.3.1 Network intrusion detection systems (NIDS)

5.3.2 Host-based intrusion detection system (HIDS)

5.3.3 Stack-based intrusion system (SIDS)

(3 × 3) (9)

5.4 Briefly explain how the following forms of network attacks occur:

5.4.1 Attacks through information leaks

5.4.2 Man-in-the-middle attacks

5.4.3 Attacks by SQL code injection

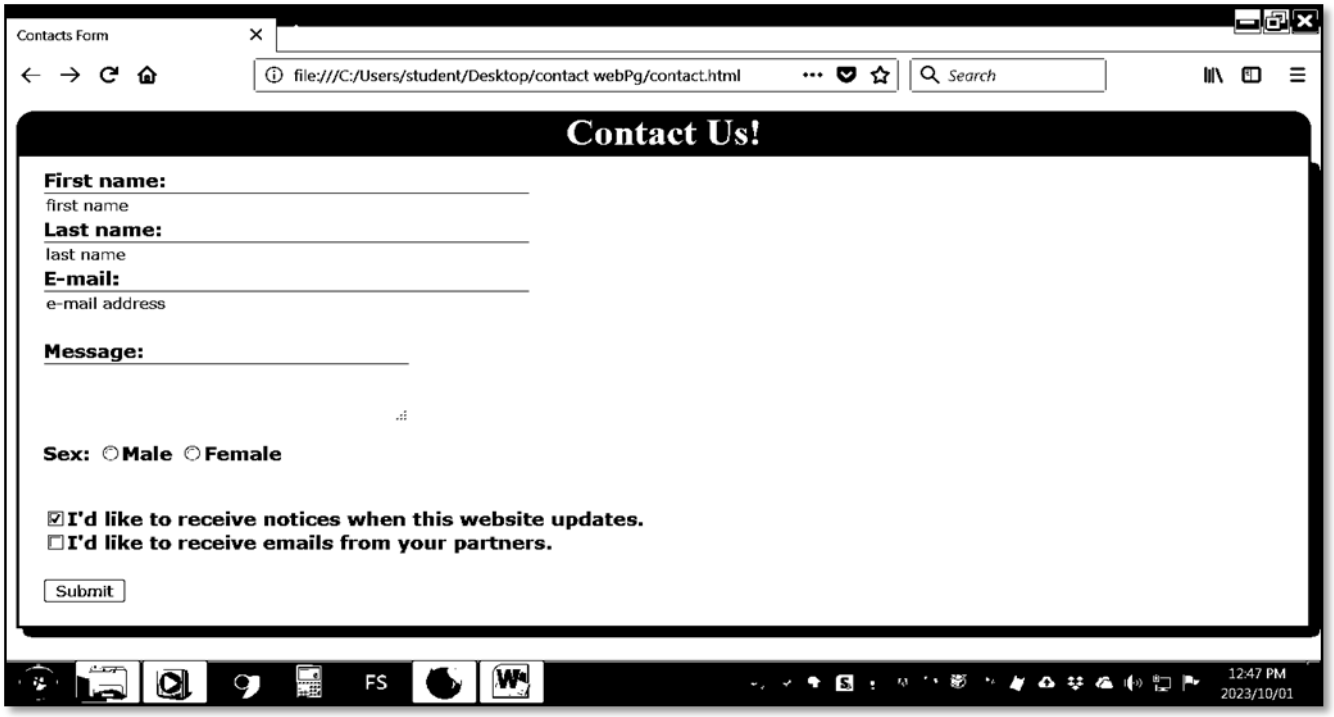
(3 × 2) (6)

[25]

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QUESTION 6

Study the following webpage and answer the questions.



6.1 Produce the complete HTML code for this web site shown above. (12)

6.2 Complete the CSS code for aligning the header as relative, background as yellow, and height as 40px. (3)

6.3 Produce the CSS code to format the text-area as follows (vertical-align = top, width = 300px and color = red). (3)

6.4 Indentify THREE types of CSS that could be used to control a web page. (3)

6.5 Name FOUR browsers that could be used to view web pages. (4)
[25]

TOTAL SECTION B: 125
GRAND TOTAL: 150

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